Overview of Hidden Markov model

Subject: Hidden Markov model

$$P(O \mid \lambda) = \sum_{Q} P(O \mid Q, \lambda) P(Q \mid \lambda)$$

1.

Computational finance Single-molecule kinetic analysis Neuroscience Cryptanalysis Speech recognition, including Siri Speech synthesis Part-of-speech tagging Document separation in scanning solutions Machine translation Partial discharge Gene prediction Handwriting recognition Alignment of bio-sequences Time series analysis Activity recognition Protein folding Sequence classification Metamorphic virus detection Sequence motif discovery (DNA and proteins) DNA hybridization kinetics Chromatin state discovery Transportation forecasting Solar irradiance variability

2.

Probability of an observed sequence The task is to compute in a best way, given the parameters of the model, the probability of a particular output sequence. This requires summation over all possible state sequences: The probability of observing a sequence

3.

In this piece of code, start_probability represents Alice's belief about which state the HMM is in when Bob first calls her (all she knows is that it tends to be rainy on average). The particular probability distribution used here is not the equilibrium one, which is (given the transition probabilities) approximately {'Rainy': 0.57, 'Sunny': 0.43}. The transition_probability represents the change of the weather in the underlying Markov chain. In this example, there is only a 30% chance that tomorrow will be sunny if today is rainy. The emission_probability represents how likely Bob is to perform a certain activity on each day. If it is rainy, there is a 50% chance that he is cleaning his apartment; if it is sunny, there is a 60% chance that he is outside for a walk.

4.

Most likely explanation The task, unlike the previous two, asks about the joint probability of the entire sequence of hidden states that generated a particular sequence of observations (see illustration on the right). This task is generally applicable when HMM's are applied to different sorts of problems from those for which the tasks of filtering and smoothing are applicable. An example is part-of-speech tagging, where the hidden states represent the underlying parts of speech corresponding to an observed sequence of words. In this case, what is of interest is the entire sequence of parts of speech, rather than simply the part of speech for a single word, as filtering or smoothing would compute. This task requires finding a maximum over all possible state sequences, and can be solved efficiently by the Viterbi algorithm.

5.

Given a Markov transition matrix and an invariant distribution on the states, a probability measure can be imposed on the set of subshifts. For example, consider the Markov chain given on the right on the states A , B 1 , B 2 {\displaystyle A,B_{1},B_{2}} , with invariant distribution π = (2 / 7 , 4 / 7 , 1 / 7) {\displaystyle \pi =(2/7,4/7,1/7)} . By ignoring the distinction between B 1 , B 2 {\displaystyle B_{1},B_{2}} , this space of subshifts is projected on A , B 1 , B 2 {\displaystyle A,B_{1},B_{2}} into another space of subshifts on A , B {\displaystyle A,B} , and this projection also projects the probability measure down to a probability measure on the subshifts on A , B {\displaystyle A,B} . The curious thing is that the probability measure on the subshifts on A , B {\displaystyle A,B} is not created by a Markov chain on A , B {\displaystyle A,B} , not even multiple orders. Intuitively, this is because if one observes a long sequence of B n {\displaystyle B^{n}} , then one would become increasingly sure that the Pr (A ■ B n) → 2 3 {\displaystyle \Pr(A\mid B^{n})\to {\frac {2}{3}}} , meaning that the observable part of the system can be affected by something infinitely in the past. Conversely, there exists a space of subshifts on 6 symbols, projected to subshifts on 2 symbols, such that any Markov measure on the smaller subshift has a preimage measure that is not Markov of any order (example 2.6).

6.

History Hidden Markov models were described in a series of statistical papers by Leonard E. Baum and other authors in the second half of the 1960s. One of the first applications of HMMs was speech recognition, starting in the mid-1970s. From the linguistics point of view, hidden Markov models are equivalent to stochastic regular grammar. In the second half of the 1980s, HMMs began to be applied to the analysis of biological sequences, in particular DNA. Since then, they have become ubiquitous in the field of bioinformatics.

7.

Other extensions Yet another variant is the factorial hidden Markov model, which allows for a single observation to be conditioned on the corresponding hidden variables of a set of K {\displaystyle K} independent Markov chains, rather than a single Markov chain. It is equivalent to a single HMM, with N K {\displaystyle N^{K}} states (assuming there are N {\displaystyle N} states for each chain), and therefore, learning in such a model is difficult: for a sequence of length T {\displaystyle T} , a straightforward Viterbi algorithm has complexity O (N 2 K T) {\displaystyle O(N^{2K}\,T)} . To find an exact solution, a junction tree algorithm could be used, but it results in an O (N K + 1 K T) {\displaystyle O(N^{K+1}\,K\,T)} complexity. In practice, approximate techniques, such as variational approaches, could be used. All of the above models can be extended to allow for more distant dependencies among hidden states, e.g. allowing for a given state to be dependent on the previous two or three states rather than a single previous state; i.e. the transition probabilities are extended to encompass sets of three or four adjacent states (or in general K {\displaystyle K} adjacent states). The disadvantage of such models is that dynamic-programming algorithms for training them have an O (N K T) {\displaystyle O(N^{K}\,T)} running time, for K {\displaystyle K} adjacent states and T {\displaystyle T} total observations (i.e. a length- T {\displaystyle T} Markov chain). This extension has been widely used in bioinformatics, in the modeling of DNA sequences. Another recent extension is the triplet Markov model, in which an auxiliary underlying process is added to model some data specificities. Many variants of this model have been proposed. One should also mention the interesting link that has been established between the theory of evidence and the triplet Markov models and which allows to fuse data in Markovian context and to model nonstationary data. Alternative multi-stream data fusion strategies have also been proposed in recent literature, e.g., Finally, a different rationale towards addressing the problem of modeling nonstationary data by means of hidden Markov models was suggested in 2012. It consists in employing a small recurrent neural network (RNN), specifically a reservoir network, to capture the evolution of the temporal dynamics in the observed data. This information, encoded in the form of a high-dimensional vector, is used as a conditioning variable of the HMM state transition probabilities. Under such a setup, eventually is obtained a nonstationary HMM, the transition probabilities of which evolve over time in a manner that is inferred from the data, in contrast to some unrealistic ad-hoc model of temporal evolution. In 2023, two innovative algorithms were introduced for the Hidden Markov Model. These algorithms enable the computation of the posterior distribution of the HMM without the necessity of explicitly modeling the joint distribution, utilizing only the conditional distributions. Unlike traditional methods such as the Forward-Backward and Viterbi algorithms, which require knowledge of the joint law of the HMM and can be computationally intensive to learn, the Discriminative Forward-Backward and Discriminative Viterbi algorithms circumvent the need for the observation's law. This breakthrough allows the HMM to be applied as a discriminative model, offering a more efficient and versatile approach to leveraging Hidden Markov Models in various applications. The model suitable in the context of longitudinal data is named latent Markov model. The basic version of this model has been extended to include individual covariates, random effects and to model more complex data structures such as multilevel data. A complete overview of the latent Markov models, with special attention to the model assumptions and to their practical use is provided in